

Control Structures

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Introduction

In this lab you will implement a simple calculator on the target hardware. The operands can be entered through the dip switches and the operation can be selected through the rotary switch.

Learning Objectives

- you can implement a switch-case statement with a jump table in assembly
- you strengthen your skills in the application of arithmetic and logic instructions

Task 1

Implement a program, which reads 8-bit values from the DIP-switches and performs different operations on them. The result shall be displayed on the LEDs. Use the given project frame.

- Read the comments in the given program frame. They will show you where to enter your code.
- Implement the switch-case statement based on the lecture slides.

The program shall meet the following requirements (See Table 1, Figure 1):

- The first operand (op1) shall be entered through the DIP-switches S15 to S8.
- The second operand (op2) shall be entered through the DIP-switches S7 to S0.
- The 8 bit operands shall be zero-extended to 32 bit.
- The operation shall be selected through the rotary switch, according to Table 1.
- The position of the rotary switch shall be displayed on the 7-segment display. The operands 1 and 2 shall be displayed on LED15 to LED0 (above the corresponding DIP-switches).
- The 16 bit result shall be displayed on LED31 to LED16.
- A logical “1” shall mean that the corresponding LED is on.

- The program shall use a jump table.

Hex-Switch	Operation
0x00	all 16 LEDs off
0x01	$op1 + op2$
0x02	$op1 - op2$
0x03	$op1 * op2$ (unsigned)
0x04	$op1 \& op2$ > AND
0x05	$op1 op2$ > OR
0x06	$op1 \wedge op2$ > XOR
0x07	$!op1$ > NOT
0x08	$!(op1 \& op2)$ > NAND
0x09	$!(op1$
0x0A	$!(op1 \wedge op2)$ > XNOR
0x0B	all 16 LEDs on
0x0C	all 16 LEDs on
0x0D	all 16 LEDs on
0x0E	all 16 LEDs on
0x0F	all 16 LEDs on

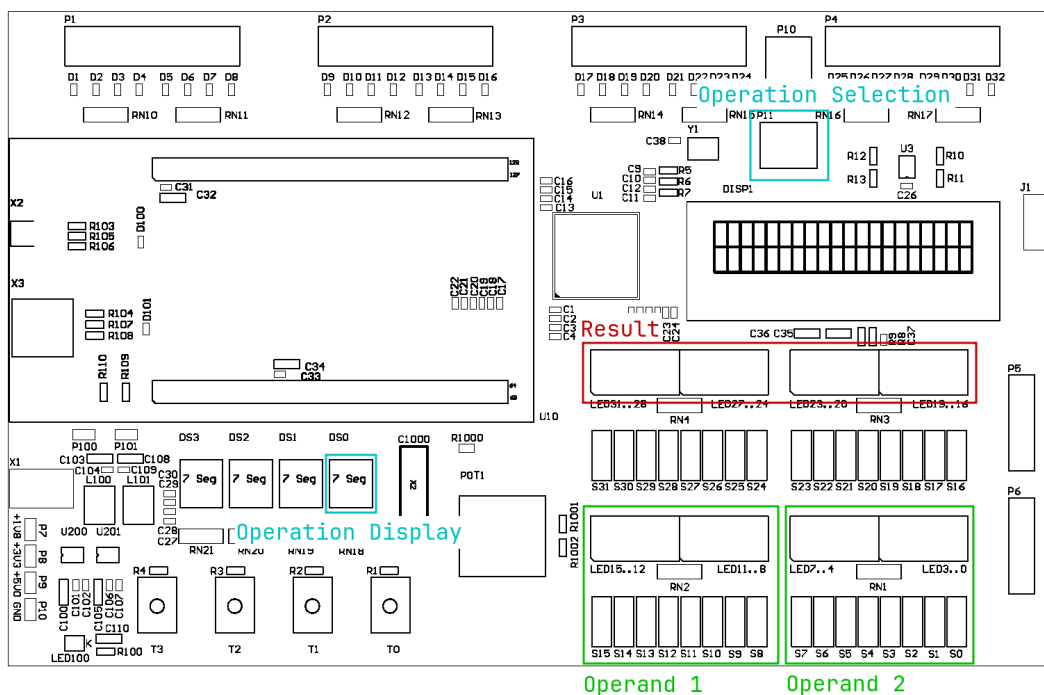


Figure 1: in- and output of operands, operation and result

Grading

The working program has to be presented to the lecturer. The student has to understand the solution/source code and has to be able to explain it to the lecturer.

Criteria	Weight
The program meets the required functionality.	4 / 4